AI Assistant for Vietnamese Labor Law: A Case Study on Employment Contract Termination

Thesis Report

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Abstract

TODO: Write a concise 200–300 word abstract after completing the evaluation. Mention Vietnamese labor law QA, RAG, graph-augmented retrieval, Neo4j, legal citations, and the main comparative findings.

Acknowledgements

TODO: Add acknowledgements before submission.

Contents

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Chapter 1

Introduction

1.1 Background

TODO: Introduce Vietnamese labor law as a high-stakes information access domain. Explain why legal texts are hard for non-experts to search, interpret, and cite correctly.

1.2 Problem Statement

TODO: State the limitations of keyword search and retrieval-only RAG for structured legal questions, multi-hop legal references, and citation-grounded answers.

1.3 Objectives

TODO: Describe objectives such as building a legal QA pipeline, comparing retrieval-only RAG and graph-augmented RAG, supporting legal citations, and evaluating answer quality.

1.4 Scope of the Project

TODO: Specify covered legal sources, Vietnamese labor law focus, supported question types, and exclusions such as formal legal advice or exhaustive statutory coverage.

1.5 Contributions

TODO: List contributions: hierarchical legal chunking, metadata-enriched retrieval, Neo4j graph modeling, query routing, citation-aware generation, tracing, and evaluation design.

1.6 Thesis Structure

TODO: Provide a paragraph mapping Chapters 2–7 to background, methodology, implementation, evaluation, discussion, and conclusion.

Chapter 2

Background and Related Work

2.1 Large Language Models

TODO: Introduce LLM capabilities and limitations, especially hallucination, context window limits, and domain-specific legal reasoning constraints.

2.2 Retrieval-Augmented Generation

TODO: Explain indexing, embedding, retrieval, context construction, and generation. Discuss why RAG is suitable for legal QA. Add citations such as [1].

Placeholder figure

Standard RAG pipeline: user query, embedding, vector search, retrieved context, prompt assembly, LLM answer.

Figure 2.1: Standard retrieval-augmented generation pipeline.

2.3 Legal Question Answering

TODO: Discuss legal domain constraints: traceability, citation correctness, jurisdiction specificity, temporal validity, and careful wording.

2.4 Graph-Augmented Retrieval

TODO: Explain how graphs can represent legal hierarchy and cross-references. Compare graph traversal with vector similarity retrieval.

2.5 Evaluation Methods for RAG Systems

TODO: Summarize metrics such as Recall@k, Precision@k, MRR, faithfulness, answer relevance, context precision, context recall, and citation matching accuracy.

Chapter 3

System Design and Methodology

3.1 Overall System Architecture

TODO: Describe data ingestion, preprocessing, vector indexing, graph construction, query routing, retrieval, generation, citation validation, tracing, and user interface layers.

Placeholder figure

Overall architecture: legal corpus, preprocessing, vector database, Neo4j, query router, retrieval pipelines, LLM, observability, frontend.

Figure 3.1: Overall system architecture.

3.2 Data Sources

TODO: List official or curated legal sources, document versions, collection method, update date, and licensing or access notes.

Table 3.1: Data sources used by the system.

Source	Document Type	Coverage	Status
TODO	Labor Code	TBD	To be filled
TODO	Decree or Circular	TBD	To be filled
TODO	Guidance document	TBD	To be filled

3.3 Legal Document Preprocessing

TODO: Describe text extraction, normalization, structural parsing, cleaning, deduplication, metadata assignment, and validation.

3.3.1 Hierarchical Chunking for Legal Texts

TODO: Explain the Document -¿ Chapter -¿ Section -¿ Article -¿ Clause -¿ Point hierarchy and why preserving this structure improves retrieval and citation grounding.

3.3.2 Metadata Enrichment

TODO: Define fields such as document title, article number, clause number, point label, effective date, parent path, citation string, and source URI.

3.3.3 Rechunking Strategy

TODO: Describe when adjacent units are merged, when long provisions are split, and how parent-child metadata is retained after rechunking.

3.4 Vector-based Retrieval

TODO: Specify embedding model, vector store, similarity metric, top-k setting, filters, reranking if used, and context assembly policy.

3.5 Graph Construction with Neo4j

TODO: Define node labels, relationship types, uniqueness constraints, and how legal hierarchy and references are inserted into Neo4j.

Placeholder figure

Neo4j graph schema: Document, Chapter, Section, Article, Clause, Point, Concept, and reference relationships.

Figure 3.2: Neo4j graph schema for hierarchical legal documents.

3.6 Query Routing and Retrieval Strategy

TODO: Describe the query router inputs, outputs, confidence scores, fallback behavior, and how routing affects retrieval mode.

3.6.1 Query Intent Classification

TODO: Describe possible intents such as definition lookup, article-specific lookup, eligibility question, comparison question, procedure question, and multi-hop reasoning question.

3.6.2 Retrieval Strategy Selection

TODO: Explain when to use vector retrieval, graph traversal, hybrid retrieval, metadata filtering, or reranking.

3.7 Graph-Augmented Retrieval Pipeline

TODO: Explain how initial candidates seed graph expansion, how neighboring legal units are selected, and how graph context is merged with vector context.

Placeholder figure

Graph-augmented retrieval flow: query, intent, seed retrieval, graph traversal, context merge, citation-aware generation.

Figure 3.3: Graph-augmented retrieval flow.

3.8 Answer Generation with Legal Citations

TODO: Describe answer constraints, citation insertion, unsupported-answer handling, and post-generation citation validation.

Chapter 4

Implementation

4.1 Technology Stack

TODO: Describe the backend, frontend, vector database, Neo4j, LLM provider, tracing tools, evaluation scripts, and deployment environment.

Table 4.1: Technology stack.

Layer	Technology	Purpose
Backend	TODO	API, orchestration, retrieval, generation
Frontend	TODO	User interface for legal QA
Vector database	TODO	Embedding search baseline
Graph database	Neo4j	Legal hierarchy and graph traversal
Evaluation	TODO	Benchmarking and report generation

4.2 Backend Implementation

TODO: Explain request flow, service modules, retrieval interfaces, generator interface, error handling, and configuration management.

4.3 Frontend Implementation

TODO: Describe the question input, answer display, citation presentation, trace or debug view, and user feedback collection if implemented.

4.4 Vector Database Implementation

TODO: Explain collection schema, embedding storage, metadata filters, similarity search, and update or rebuild process.

4.5 Neo4j Graph Database Design

TODO: Document constraints, indexes, Cypher queries, import scripts, relationship creation, and graph traversal patterns.

4.6 Prompt Design

TODO: Explain system messages, answer style constraints, citation requirements, refusal policy for unsupported questions, and prompt iteration process.

```
System:
You are a Vietnamese labor law assistant. Answer only from the provided legal
    context.
If the context is insufficient, say that the answer cannot be determined from
    the provided sources.

User question:
{question}

Retrieved legal context:
{context}

Instructions:
1. Provide a concise answer in English or Vietnamese according to the user
    request.
2. Cite each legal claim using the supplied citation identifiers.
3. Do not invent articles, clauses, or legal consequences.
4. End with a short list of cited sources.
```

Listing 4.1: Prompt template placeholder for citation-aware legal answering.

4.7 Example Use Cases

TODO: Describe example questions such as severance allowance, probation period, overtime limits, contract termination, and article-specific lookup.

4.8 System Observability and Tracing

TODO: Document trace fields such as query ID, routing decision, retrieved chunk IDs, graph traversal path, prompt size, latency, token usage, and citation validation status.

4.9 Deployment Setup and CI/CD

TODO: Explain local development setup, Docker services, environment variables, database migrations or imports, CI checks, and deployment target.

Chapter 5

Evaluation and Results

5.1 Evaluation Setup

TODO: State hardware, software versions, corpus snapshot, embedding model, LLM model, retrieval settings, graph settings, and random seeds where applicable.

Placeholder figure

Evaluation pipeline: benchmark questions, retrieval runs, generation runs, metric computation, LLM-as-Judge review, citation matching, reporting.

Figure 5.1: Evaluation pipeline.

5.2 Curated Legal QA Benchmark Dataset

TODO: Explain question selection, answer references, citation labels, quality control, and train/test separation if any.

Table 5.1: Benchmark question types.

Question Type	Description	Count
Definition lookup	TODO	TBD
Article-specific lookup	TODO	TBD
Multi-hop legal reasoning	TODO	TBD
Procedure or entitlement	TODO	TBD

5.3 Retrieval Evaluation Metrics

TODO: Define Recall@k, Precision@k, Mean Reciprocal Rank, nDCG if used, and how relevant legal chunks are determined.

Table 5.2: Retrieval metrics.

Metric	Definition	Notes
Recall@k	TODO	Measures whether relevant chunks appear in top-k
Precision@k	TODO	Measures relevance density in retrieved context
MRR	TODO	Measures rank of first relevant result
nDCG@k	TODO	Optional graded relevance metric

5.4 Answer Quality Evaluation

TODO: Describe human review, LLM-as-Judge, RAGAS-style metrics, scoring rubrics, and safeguards against evaluator bias.

5.4.1 Faithfulness

TODO: Explain how unsupported claims are identified and scored against retrieved context and cited legal provisions.

5.4.2 Answer Relevance

TODO: Explain how directly and completely an answer responds to the user question.

5.4.3 Context Precision and Context Recall

TODO: Explain how retrieved contexts are judged for usefulness and coverage of required legal evidence.

Table 5.3: Answer quality metrics.

Metric	Scoring Method	Status
Faithfulness	TODO	To be filled after evaluation
Answer relevance	TODO	To be filled after evaluation
Context precision	TODO	To be filled after evaluation
Context recall	TODO	To be filled after evaluation

5.5 Citation Evaluation

TODO: Explain exact, partial, and incorrect citation matches. Define how hallucinated or missing citations are counted.

5.5.1 Citation Matching Accuracy

TODO: Use this formula and later report the measured score after running evaluation.

$$\text{Citation Matching Accuracy} = \frac{\text{Number of answers with correct citations}}{\text{Total number of evaluated answers}} \tag{5.1}$$

5.5.2 Citation Hallucination Analysis

TODO: Categorize hallucinations such as nonexistent articles, wrong clauses, cited but unused evidence, and unsupported answer claims.

Table 5.4: Citation evaluation metrics.

Metric	Definition	Value
Citation matching accuracy	See Equation ??	TBD
Missing citation rate	TODO	TBD
Citation hallucination rate	TODO	TBD
Wrong-granularity citation rate	TODO	TBD

5.6 Experimental Results

TODO: Fill in this section only after running experiments. Do not report speculative improvements.

Table 5.5: Main experimental results.

System	Recall@k	Faithfulness	Citation Accuracy	Latency
Retrieval-only RAG	TBD	TBD	TBD	TBD
Graph-augmented RAG	TBD	TBD	TBD	TBD

5.7 Comparison between Retrieval-only RAG and Graph-Augmented RAG

TODO: Discuss where graph augmentation improves retrieval, where it does not, and how it affects answer quality, citation correctness, and latency.

5.8 Error Analysis

TODO: Group errors by retrieval miss, routing error, graph traversal error, generation error, citation error, and ambiguous legal question.

Table 5.6: Error analysis examples.

Error Type	Example Question	Observed Behavior	Planned Fix
Retrieval miss	TODO	TBD	TBD
Citation error	TODO	TBD	TBD
Routing error	TODO	TBD	TBD

5.9 Case Studies

TODO: Choose representative questions that show strengths and limitations of both retrieval-only and graph-augmented RAG.

Chapter 6

Discussion

6.1 Key Findings

TODO: Discuss the most important findings supported by the experimental results and case studies.

6.2 Impact of Hierarchical Chunking

TODO: Explain whether the Document -¿ Chapter -¿ Section -¿ Article -¿ Clause -¿ Point structure improved evidence localization and reduced context noise.

6.3 Benefits of Graph-Augmented Retrieval

TODO: Explain how graph traversal helped with legal hierarchy, neighboring provisions, cross-references, and multi-hop questions.

6.4 Limitations

TODO: Cover corpus coverage, model dependence, benchmark size, citation validation limits, legal interpretation risks, and operational constraints.

6.5 Practical Applications

TODO: Describe possible applications for students, HR staff, legal researchers, and public information access, while clarifying that the system should not replace professional legal advice.

Chapter 7

Conclusion and Future Work

7.1 Conclusion

TODO: Summarize the project goals, implemented system, evaluation approach, and confirmed results after experiments are complete.

7.2 Future Work

TODO: Discuss expanding the legal corpus, temporal legal versioning, stronger citation verification, human expert evaluation, multilingual support, and production monitoring.

References

[1] Patrick Lewis et al. “Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks”. In: *Advances in Neural Information Processing Systems*. 2020.

Appendix A

Supplementary Materials

A.1 Benchmark Dataset Template

TODO: Document benchmark fields such as question ID, question text, expected answer, gold citations, question type, and difficulty.

A.2 Prompt Templates

TODO: Include the final system prompt, retrieval prompt, router prompt, and evaluator prompt after they are stabilized.

A.3 Configuration Details

TODO: List key environment variables, retrieval parameters, model versions, database versions, and evaluation script commands.